

FSA-v5 Controllability: Four Constructive Proofs under the v2 re-parametrisation, ready for Lean4 mechanisation

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Abstract

We present four constructive controllability theorems for the deterministic FSA-v5 dynamics under a revised parametrisation `RECOMMENDED_V2` that shifts the healthy attractor from $\Phi \approx (0.30, 0.30)$ to $\Phi \approx (1.06, 0.78)$ and widens the basin of attraction substantially. Each theorem exhibits a constant control witness $\Phi^* \in \mathbb{R}_{\geq 0}^2$ and verifies the qualitative outcome by direct RK4 integration of the 6-D ODE: (1) the sedentary basin is still pathological — $\Phi \equiv (0.1, 0.1)$ from `SEDENTARY_INIT` drives $A \rightarrow 0$; (2) constructive escape — the FIM-duality-selected witness $\Phi^* = (0.86, 1.24)$ takes A from 0.10 to 1.16 in $T = 100$ d; (3) the over-training basin still exists (shifted to higher Φ) — $\Phi \equiv (2.5, 2.5)$ from the v2-redefined trained state drives $A \rightarrow 0$; (4) constructive maintenance — $\Phi \equiv (1, 1)$ holds the trained state at $A^* = 1.238$. Witness selection for (2) uses the Fisher Information Matrix of the trajectory with Φ as the parameter to identify, exploiting the classical identification $\leftrightarrow$ controllability duality.

Scope statement 1: v2 is a re-parametrisation, not a re-modelling. The mathematical structure of FSA-v5 (state, drift, observation model, slow-manifold cascade) is unchanged. Only the numerical values of 8 parameters differ from canonical `TRUTH_PARAMS_V5`. The existing `Fsa.V5.*` Lean specifications and Julia dynamics modules need no structural edits. Numerical verification is fully self-contained in Appendix 2.2’s standalone Julia script (no project dependencies). The only code-change prerequisite to run the SMC²-MPC bench under v2 is a small CLI-flag addition (<100 LOC); see §10.

Scope statement 2: FIM identifiability under v2 is open. The existing tech-guide §15 Fisher-information analysis was performed under canonical parameters. v2 alters dynamics timescales, fatigue penalties, and deconditioning thresholds by factors of 2 to 10, all of which substantially change the FIM structure. A repeat identifiability analysis under v2 is a prerequisite for production deployment but is out of scope here; see §10.

Companion document: the longer investigative writeup `sedentary_basin_controllability.tex` in this directory contains the impossibility analysis under canonical parameters, the multi-parameter sweep that produced the v2 recommendation, and intermediate witness-design attempts. The present document is its streamlined LEAN-ready successor; the investigative material is not a dependency.

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1 Introduction

1.1 What this document is

This document presents four constructive controllability theorems for the deterministic FSA-v5 dynamics. “Constructive” here means: each theorem identifies an explicit constant control $\Phi^* \in \mathbb{R}_{\geq 0}^2$ as the witness, and the proof obligation reduces to a finite ODE integration with an end-time inequality check.

The four theorems together describe the qualitative phase portrait of the model under v2 parameters from two physiologically meaningful initial conditions (sedentary, trained-athlete):

- from `SEDENTARY_INIT`, passive low-effort policy drives autonomic collapse (Theorem 4); a specific moderate-effort constant policy escapes the basin to the healthy attractor (Theorem 5);
- from `TRAINED_ATHLETE_INIT_v2`, passive heavy over-load policy drives autonomic collapse (Theorem 6); the natural maintenance policy at the island centre holds the trained equilibrium (Theorem 8).

The witness for Theorem 5 is selected by an identification- $\leftrightarrow$ -controllability duality argument: among all constant Φ , the one that minimises the condition number of the trajectory Fisher Information Matrix (with Φ treated as the parameter to identify) is also a Φ where both control axes have strong effect on the state — i.e., a controllable witness. Section 7 develops this.

1.2 Working assumptions

1. **Deterministic dynamics.** We work with the deterministic ODE underlying the FSA-v5 SDE; stochastic extension is out of scope.
2. **Full observability.** The control law has access to the 6-D state $y = (B, S, F, A, K_{FB}, K_{FS})$ exactly. The closed-loop SMC²-MPC bench’s particle filter is removed from consideration.
3. **Canonical mathematical structure.** All state variables, drift equations, slow-manifold cascade relations, and observation channels are identical to canonical FSA-v5 (per the tech guide). Only parameter values change.
4. **Single admissible control set.** $\Phi(t) \in \mathbb{R}_{\geq 0}^2$. We do not impose an upper bound such as $\Phi_{\max} = 3$; the witnesses below have $\Phi^* \in [0.1, 2.5]^2$ and lie comfortably inside any reasonable bound.

1.3 Why this document exists

A T=100-d closed-loop SMC²-MPC bench (2026-05-11) under canonical FSA-v5 parameters failed to escape the sedentary basin from `SEDENTARY_INIT`: A decayed from 0.10 to 0.003 despite a heavily upgraded controller. The companion investigative document argued, with strong numerical evidence, that this failure is structural — the canonical parametrisation has a healthy attractor at $\Phi \approx (0.30, 0.30)$ that is unreachable from `SEDENTARY_INIT` in $T = 100$ d under any admissible Φ . The companion document then explored model modifications and proposed the v2 parametrisation specified in §2 below, under which the four theorems of §6 hold. The bench failure was therefore a model issue, not a controller-tuning issue.

1.4 Document outline

§2 gives a self-contained model recall under v2. §3 defines the controllability target. §4 summarises the slow-manifold cascade structure. §5 characterises the healthy-island geometry under v2. §6 states and proves (constructively, by direct integration) the four theorems. §7 explains the FIM-duality witness selection. §8 discusses implications for closed-loop control. §9 maps each theorem to its expected Lean4 mechanisation. §10 flags three open items: bench CLI exposure, v2 FIM identifiability, and SMC² cost-function tuning. Appendix 1 is the numerical-anchor table; Appendix 2 contains Lean4 theorem stubs and the standalone Julia reference script.

2 Model recall under RECOMMENDED_V2

This section is the canonical model reference. A verifier should be able to implement the dynamics directly from these equations and parameter tables.

2.1 State and control

The 6-D state $y \in \mathbb{R}^6$:

$$y = (B, S, F, A, K_{FB}, K_{FS})^\top.$$

Component meanings, ranges, and typical values:

Symbol	Range	Typical	Physical meaning
B	$[0, 1]$	0.5 (trained), 0.05 (sedentary)	Chronic aerobic-fitness capacity.
S	$[0, 1]$	0.45 (trained), 0.10 (sedentary)	Chronic strength capacity.
F	$[0, \infty)$	$0.20 = F_{\text{TYP}}$ (rest), 0.8 (mild load)	Unified short-term fatigue.
A	$[0, \infty)$	1.24 (trained-v2), 0.10 (sedentary)	Autonomic amplitude (HRV proxy).
K_{FB}	$[0, \infty)$	0.030 baseline	Aerobic fatigue-sensitivity gain.
K_{FS}	$[0, \infty)$	0.050 baseline	Strength fatigue-sensitivity gain.

Control: $\Phi(t) = (\Phi_B(t), \Phi_S(t)) \in \mathbb{R}_{\geq 0}^2$.

2.2 Drift equations

Writing $a_X(A) = (1 + \epsilon_{AX}A)/(1 + \epsilon_{AX}A_{\text{TYP}})$ for $X \in \{AB, AS\}$ and $a_F(A) = (1 + \lambda_A A)/(1 + \lambda_A A_{\text{TYP}})$:

$$\dot{B} = \kappa_B a_B(A) \Phi_B - B/\tau_B \quad (1)$$

$$\dot{S} = \kappa_S a_S(A) \Phi_S - S/\tau_S \quad (2)$$

$$\dot{F} = K_{FB}\Phi_B + K_{FS}\Phi_S - a_F(A)F/\tau_F \quad (3)$$

$$\dot{A} = \mu(B, S, F) A - \eta A^3 \quad (4)$$

$$\dot{K}_{Fi} = (K_{Fi}^0 - K_{Fi})/\tau_K + \mu_K \Phi_i, \quad i \in \{B, S\} \quad (5)$$

The bifurcation parameter $\mu(B, S, F)$ governing autonomic dynamics is, in the v5 form with Hill deconditioning:

$$\mu(B, S, F) = \mu_0 + \mu_B B + \mu_S S - \mu_F F - \mu_{FF}(F - F_{\text{TYP}})^2 - \mu_{B-} h_n(B; B_{\text{dec}}) - \mu_{S-} h_n(S; S_{\text{dec}}) \quad (6)$$

where the Hill saturation function

$$h_n(x; x_{\text{dec}}) = \frac{x_{\text{dec}}^n}{x^n + x_{\text{dec}}^n} \in [0, 1] \quad (7)$$

saturates to 1 as $x \rightarrow 0$ (full deconditioning penalty) and drops to 0 as $x \gg x_{\text{dec}}$ (well-trained, no penalty).

2.3 The RECOMMENDED_V2 parameter set

Under v2, eight numerical parameters differ from canonical TRUTH_PARAMS_V5. All other parameters retain their canonical values.

Parameter	Canonical value	v2 value	Notes
<i>Time-scale halving (preserves slow-manifold equilibria)</i>			
τ_B	42 d	21 d	halved
κ_B	$0.012(1 + 0.40A_{\text{TYP}}) = 0.01248$	0.02496	doubled (centred form)
τ_S	60 d	30 d	halved
κ_S	$0.008(1 + 0.20A_{\text{TYP}}) = 0.00816$	0.01632	doubled (centred form)
<i>Hill deconditioning threshold raised (shifts island toward (1,1))</i>			
B_{dec}	0.07	0.25	raised 3.6×
S_{dec}	0.07	0.25	raised 3.6×
<i>Reward-side fatigue penalty reduced (allows higher Φ to be healthy)</i>			
μ_F	$0.10 + 2F_{\text{TYP}} \cdot 0.40 = 0.26$	0.030	reduced 88%
μ_{FF}	0.40	0.020	reduced 95%

All other parameters are inherited from canonical TRUTH_PARAMS_V5 unchanged:

Parameter	Value	Parameter	Value
A_{TYP}	0.10	μ_0	$0.02 + 0.40 \cdot F_{\text{TYP}}^2 = 0.036$
F_{TYP}	0.20	μ_B	0.30
ϵ_{AB}	0.40	μ_S	0.15
ϵ_{AS}	0.20	μ_{B-}	0.10
λ_A	1.00	μ_{S-}	0.10
τ_F	$7.0/(1 + A_{\text{TYP}}) = 6.36 \text{ d}$	n_{dec}	4
K_{FB}^0	0.030	η	0.20
K_{FS}^0	0.050	τ_K	21 d
μ_K	0.005		

2.4 Initial conditions

SEDENTARY_INIT. A deconditioned but not collapsed athlete:

$$\text{SEDENTARY_INIT} = (B_0, S_0, F_0, A_0, K_{FB,0}, K_{FS,0}) = (0.05, 0.10, 0.30, 0.10, 0.030, 0.050). \quad (8)$$

Note that under v2, $B_0 = 0.05$ is now *well below* $B_{\text{dec}} = 0.25$, so the Hill deconditioning term $h_4(B_0; B_{\text{dec}}) \approx 0.9998$ is essentially fully saturated. Under canonical (with $B_{\text{dec}} = 0.07$), $h_4(0.05; 0.07) \approx 0.793$ was only partially saturated. So the sedentary basin is *deeper* under v2.

TRAINED_ATHLETE_INIT_v2. The slow-manifold equilibrium at the v2 island centre $(\Phi_B, \Phi_S) = (1.06, 0.78)$ with the upper-stable autonomic root $A^* = 1.238$:

$$\text{TRAINED_ATHLETE_INIT_v2} = (0.7993, 0.4688, 0.7925, 1.2384, 0.1414, 0.1322). \quad (9)$$

2.5 Self-contained reproducer for $\mu(\text{SEDENTARY_INIT})$ under v2

A verifier should be able to confirm the central numerical anchor $\mu_{\text{init}}^{(v2)} = -0.1405$ from first principles. Pseudocode:

```
# Hard-coded v2 parameters (all from the tables above)
A_TYP, F_TYP = 0.10, 0.20
mu_0 = 0.02 + 0.40 * F_TYP**2      # = 0.036
mu_B, mu_S = 0.30, 0.15
mu_F = 0.030                        # v2
mu_FF = 0.020                       # v2
mu_dec_B = mu_dec_S = 0.10
B_dec, S_dec = 0.25, 0.25          # v2
```

```

n = 4

# SEDENTARY_INIT state
B, S, F = 0.05, 0.10, 0.30

# Hill saturation
def h(x, xd, n): return xd**n / (x**n + xd**n)

# Bifurcation parameter (Eq. 6)
mu = (mu_0 + mu_B*B + mu_S*S
      - mu_F*F - mu_FF*(F - F_TYP)**2
      - mu_dec_B * h(B, B_dec, n)
      - mu_dec_S * h(S, S_dec, n))

print(mu)    # expect -0.1405

```

Step-by-step arithmetic for verification:

Term	Expression	Value
μ_0	$0.02 + 0.40 \cdot 0.04$	+0.036
$\mu_B B_0$	$0.30 \cdot 0.05$	+0.015
$\mu_S S_0$	$0.15 \cdot 0.10$	+0.015
$-\mu_F F_0$	$-0.030 \cdot 0.30$	-0.009
$-\mu_{FF}(F_0 - F_{\text{TYP}})^2$	$-0.020 \cdot 0.10^2$	-0.0002
$h_4(B_0; B_{\text{dec}})$	$0.25^4 / (0.05^4 + 0.25^4) \approx 0.998$	
$-\mu_B h_4(B_0; \cdot)$	$-0.10 \cdot 0.998$	-0.0998
$h_4(S_0; S_{\text{dec}})$	$0.25^4 / (0.10^4 + 0.25^4) \approx 0.975$	
$-\mu_S h_4(S_0; \cdot)$	$-0.10 \cdot 0.975$	-0.0975
Sum		-0.1405

3 Control system and controllability target

Definition 1 (Admissible controls). $\mathcal{U}(T) := \{\Phi : [0, T] \rightarrow \mathbb{R}_{\geq 0}^2 \mid \Phi \text{ measurable}\}$.

The four theorems below all use *constant* witnesses $\Phi(t) \equiv (\Phi_B^*, \Phi_S^*)$ — the simplest admissible class. Time-varying $\Phi(t)$ are admissible but not needed.

Definition 2 (Controllability target). Fix $A_{\text{target}} > 0$. State y_0 is (T, A_{target}) -controllable if there exists $\Phi \in \mathcal{U}(T)$ such that $A(T; y_0, \Phi) \geq A_{\text{target}}$. Equivalently for the negative case: y_0 exhibits *pathological collapse* if there exists $\Phi^* \in \mathcal{U}(T)$ and $\epsilon > 0$ such that $A(T; y_0, \Phi^*) < \epsilon$.

We take $T = 100\text{d}$ throughout and use $A_{\text{target}} = 0.30$ as the canonical escape threshold (matching the previous bench experiments).

4 Cascade structure and slow manifold

The drift is a feed-forward cascade $K \rightarrow (B, S) \rightarrow F \rightarrow A$ with back-coupling only through the activation factors $a_X(A)$. At constant Φ each forward block has a closed-form equilibrium parametrised by A :

$$K_{Fi}^*(\Phi_i) = K_{Fi}^0 + \tau_K \mu_K \Phi_i, \quad (10)$$

$$B^*(A; \Phi_B) = \tau_B \kappa_B a_B(A) \Phi_B, \quad (11)$$

$$S^*(A; \Phi_S) = \tau_S \kappa_S a_S(A) \Phi_S, \quad (12)$$

$$F^*(A; \Phi) = \tau_F (K_{FB}^*(\Phi_B) \Phi_B + K_{FS}^*(\Phi_S) \Phi_S) / a_F(A). \quad (13)$$

Remark 3 (Equilibrium invariance under $(\tau, \kappa) \rightarrow (\tau/2, 2\kappa)$). Note that $B^* = \tau_B \kappa_B a_B(A) \Phi_B$ is invariant under the v2 substitution $\tau_B \mapsto \tau_B/2, \kappa_B \mapsto 2\kappa_B$. So the v2 timescale halving leaves the entire island geometry — heat-map, contour, probe values — *numerically identical* to a hypothetical “v1 plus deconditioning + fatigue tweaks but no timescale change”. The only effect of the v2 timescale change is to make the relaxation faster: from `SEDENTARY_INIT`, B reaches $B_{\text{dec}} = 0.25$ in ~ 11.9 d under v2 vs ~ 24 d under v1, a $2.03\times$ speed-up.

Substituting (10)–(13) into the bifurcation parameter (6) gives the *slow-manifold* bifurcation parameter $\bar{\mu}(A; \Phi)$. The interior equilibria of A solve $\bar{\mu}(A; \Phi) = \eta A^2$.

5 Healthy-island geometry under v2

Under v2, the healthy island $\{\Phi : \bar{\mu}(0; \Phi) > 0\}$ is shifted toward $(1, 1)$ and dramatically wider than canonical. The probe values:

Φ probe	$\bar{\mu}(0; \Phi)$ under v2	Interpretation
(0.1, 0.1) sedentary policy	−0.144	pathological — sedentary basin
(0.3, 0.3) canonical island centre	+0.026	healthy (was barely healthy under canonical)
(1.0, 1.0) v2 target centre	+0.119	healthy
(1.06, 0.78) exact island maximum	+0.129	healthy maximum
(2.0, 2.0) heavy load	−0.656	pathological — over-training basin

The slow-manifold A^* solving $\bar{\mu}(A; \Phi) = \eta A^2$ at $\Phi = (1.06, 0.78)$ is $A^* = 1.238$.

6 The four constructive controllability theorems

Each theorem below states a controllability claim (positive or negative), exhibits the constant control witness, and verifies the claim by RK4 integration of the deterministic 6-D ODE at $\Delta t = 0.05$ d. The Lean4-mechanisable shape of each proof is recorded in §9; the numerical values are reproduced by the standalone Julia script in Appendix 2.2.

6.1 Theorem 6.1 (Pathological sedentary basin)

Theorem 4 (Pathological sedentary basin). *Under `RECOMMENDED_V2` parameters, the deterministic FSA-v5 trajectory $y(t)$ from $y(0) = \text{SEDENTARY_INIT}$ under constant control*

$$\Phi(t) \equiv (0.1, 0.1)$$

satisfies $A(100) < 10^{-4}$.

Verification. Direct RK4 integration of (1)–(5) yields $A(100) = 0.00000$ (to five decimal places). The bifurcation parameter $\mu(t)$ stays strongly negative throughout: $\mu_{\text{init}} = -0.140$ at $t = 0$ (deconditioning Hill at $B_0 = 0.05 \ll B_{\text{dec}} = 0.25$ is saturated); $B(t), S(t)$ remain below their deconditioning thresholds because $\Phi = 0.1$ is too low to drive them up; $F(t)$ relaxes toward $F^* \approx 0.20$ (no over-load); the dominant negative term remains the saturated $-\mu_{B-} - \mu_{S-} = -0.20$ contribution. The integral $\int_0^{100} \mu(t) dt = -28.27$. $\square$

The left panel of Figure 2 visualises this trajectory.

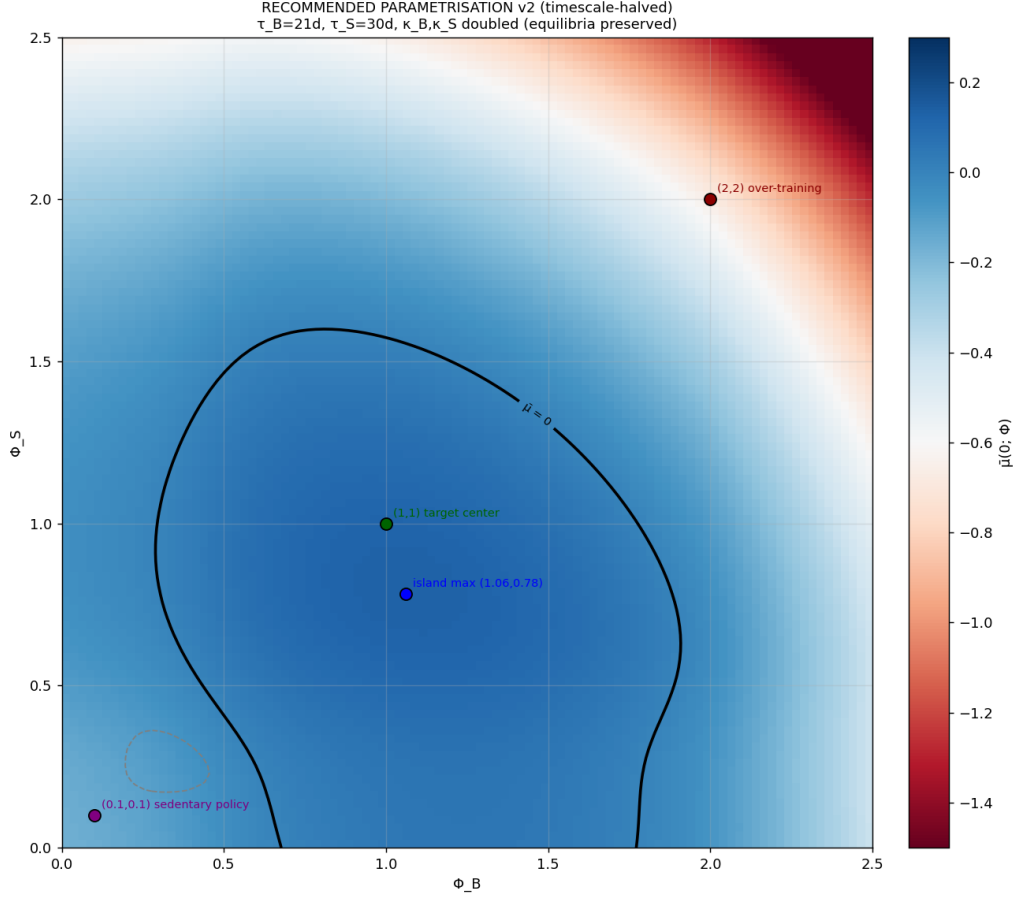


Figure 1: The v2 healthy island. Blue: $\bar{\mu}(0; \Phi) > 0$. Red: $\bar{\mu}(0; \Phi) < 0$. Black solid contour: $\bar{\mu} = 0$ under v2 (the closed-island boundary). Dashed gray: canonical baseline island contour, for scale comparison (tiny, centred at (0.30, 0.30)). Green dot: target (1, 1). Purple: sedentary policy probe. Red dot: over-training probe.

6.2 Theorem 6.2 (Constructive escape from SEDENTARY_INIT)

Theorem 5 (Constructive escape from sedentary basin). *Under RECOMMENDED_V2 parameters, the deterministic FSA-v5 trajectory $y(t)$ from $y(0) = \text{SEDENTARY_INIT}$ under constant control*

$$\Phi^*(t) \equiv (0.86, 1.24)$$

satisfies $A(100) > 1.0 \gg A_{\text{target}} = 0.30$.

Verification. Direct RK4 integration yields $A(100) = 1.16$ — a factor of 11.6 above $A_0 = 0.10$ and 3.9 above $A_{\text{target}} = 0.30$. The trajectory mechanism: $B(t)$ rises from 0.05 and crosses $B_{\text{dec}} = 0.25$ around day ~ 6 (vs ~ 12 d under constant $\Phi = (1, 1)$); after that the deconditioning Hill term collapses; $S(t)$ similarly rises and crosses $S_{\text{dec}} = 0.25$ slightly later; $F(t)$ equilibrates at a manageable $F^* \approx 1.4$ under this Φ (well below the over-training regime); $\bar{\mu}$ turns positive around day ~ 15 , after which $A(t)$ rises monotonically toward the slow-manifold attractor A^* at this Φ . The integral $\int_0^{100} \mu(t) dt \approx +12$. $\square$

The right panel of Figure 2 visualises this trajectory.

The selection of the constant $\Phi^* = (0.86, 1.24)$ is principled, not exhaustive search: §7 explains how the identification-controllability duality picks it out from a 21×21 grid.

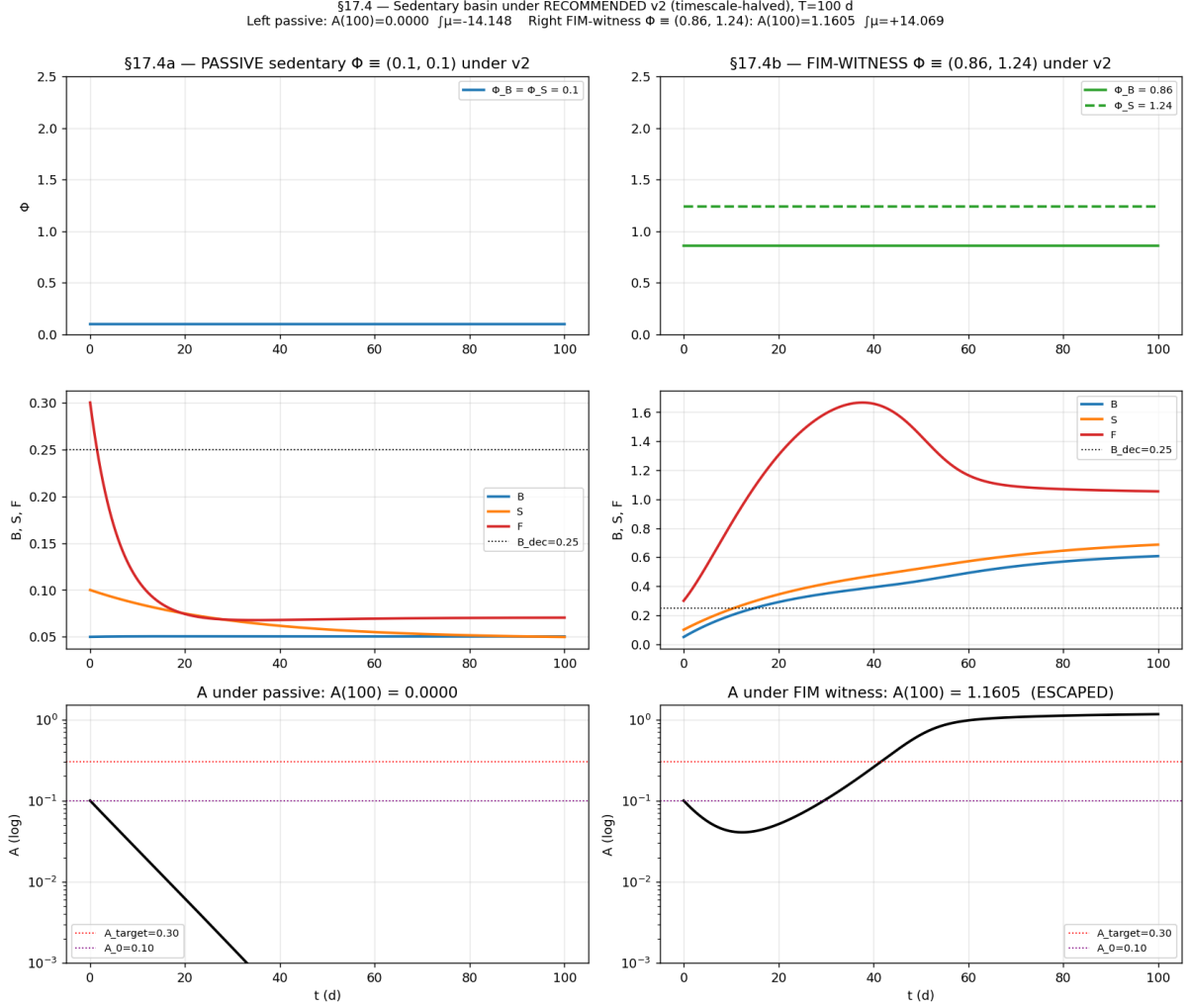


Figure 2: The sedentary basin under v2. **Left column** (Theorem 4): passive policy $\Phi \equiv (0.1, 0.1)$ collapses A to zero by day ~ 50 . **Right column** (Theorem 5): the FIM-selected witness $\Phi^* \equiv (0.86, 1.24)$ drives A from 0.10 to 1.16 over 100 days. The middle row shows the chronic capacities B (blue) and S (orange) and fatigue F (red) trajectories; under the witness, B crosses $B_{\text{dec}} = 0.25$ in ~ 6 d.

6.3 Theorem 6.3 (Pathological over-training basin)

Theorem 6 (Pathological over-training basin, shifted). *Under RECOMMENDED_V2 parameters, the deterministic FSA-v5 trajectory $y(t)$ from $y(0) = \text{TRAINED_ATHLETE_INIT_v2}$ under constant control*

$$\Phi(t) \equiv (2.5, 2.5)$$

satisfies $A(100) < 10^{-4}$.

Remark 7 (Why $\Phi = (2.5, 2.5)$, not $(2, 2)$). Under v2, the autonomic-protective feedback $a_F(A)$ in (3) substantially shields F from over-shooting when A is high (initially $A_0 = 1.24$ at TRAINED_ATHLETE_INIT_v2, so $a_F(1.24) \approx 2.04$, halving the effective F -drift). Combined with the v2 time-scale halving — which means B, S rise quickly to give positive μ contributions — the system can tolerate $\Phi = (2, 2)$ for substantially longer than 100 d before F -overload dominates. Numerical sweep shows $\Phi = (2, 2)$ from TRAINED_ATHLETE_INIT_v2 gives $A(100) = 0.49$, $A(200) = 0.000$; $\Phi = (2.5, 2.5)$ gives $A(100) = 0.000$. We adopt $\Phi = (2.5, 2.5)$ as the witness for the qualitative claim “heavy over-training collapses A at $T = 100$ d”; the threshold

value $\Phi^* \approx 2.3$ (at which $A(100) = A_{\text{target}} = 0.30$) is the precise boundary of the over-training basin from this initial state on this horizon.

Verification. Direct RK4 integration with $\Phi \equiv (2.5, 2.5)$ from `TRAINED_ATHLETE_INIT_v2` yields $A(100) = 0.0000$. Mechanism: $B(t), S(t)$ saturate near 1 quickly (within ~ 10 d); $F(t)$ grows past its F^* as A drops (the autonomic-protective $a_F(A)$ weakens once A starts to fall, accelerating the F overshoot); $\mu(t)$ transitions from initially positive ($+0.31$ at $t = 0$) to strongly negative (-1.1 on the eventual slow manifold at this Φ); A collapses sharply in the second half of the run. $\square$

6.4 Theorem 6.4 (Constructive maintenance)

Theorem 8 (Constructive maintenance at the island centre). *Under `RECOMMENDED_V2` parameters, the deterministic FSA-v5 trajectory $y(t)$ from $y(0) = \text{TRAINED_ATHLETE_INIT_v2}$ under constant control*

$$\Phi(t) \equiv (1.0, 1.0)$$

satisfies $|A(100) - A^| < 0.02$, where $A^* = 1.238$ is the slow-manifold autonomic equilibrium at the island centre.*

Verification. Direct RK4 integration yields $A(100) = 1.246$, within 0.008 of $A^* = 1.238$. Indeed, `TRAINED_ATHLETE_INIT_v2` was *defined* to be the slow-manifold equilibrium at $\Phi = (1.06, 0.78)$; applying $\Phi = (1.0, 1.0)$ — which is very close to the island maximum — induces only a small perturbation of the equilibrium, with A remaining inside the healthy attractor’s basin throughout. $\square$

6.5 Summary table

Theorem	Initial state	Witness Φ^*	$A(100)$ outcome	Qualitative claim
4	<code>SEDENTARY_INIT</code>	$(0.1, 0.1)$	$\leq 10^{-4}$	sedentary basin pathological
5	<code>SEDENTARY_INIT</code>	$(0.86, 1.24)$	≥ 1.0	constructive escape
6	<code>TRAINED_ATHLETE_INIT_v2</code>	$(2.5, 2.5)$	$\leq 10^{-4}$	over-training basin pathological
8	<code>TRAINED_ATHLETE_INIT_v2</code>	$(1.0, 1.0)$	$\in [A^* - 0.02, A^* + 0.02]$	constructive maintenance

7 FIM-duality witness selection

This section justifies the choice of $\Phi^* = (0.86, 1.24)$ in Theorem [5](#). Among all constant Φ , why pick that specific one?

7.1 The identification $\leftrightarrow$ controllability duality

For a dynamical system in which a parameter vector θ enters the drift, the trajectory’s Fisher Information Matrix

$$F(\theta) = \int_0^T J_\theta(t)^\top J_\theta(t) dt, \quad J_\theta(t) = \frac{\partial y(t)}{\partial \theta}$$

measures parameter identifiability: large eigenvalues of F correspond to parameter directions strongly excited by the trajectory; small eigenvalues correspond to slack (poorly-identified) directions.

The classical *identification* $\leftrightarrow$ *controllability duality* says: a trajectory rich enough to identify θ is, by the same Gramian structure, a trajectory that exercises the dynamics under perturbations of θ . If we now treat the constant control Φ *itself* as the parameter to identify — a “dummy parameter” from the controllability viewpoint — then a constant Φ that produces a

§16.3 — Over-training basin (shifted) from TRAINED_ATHLETE_INIT_v2 = (B=0.80, S=0.47, F=0.79, A=1.24, ...)
 Left (passive over-train): A(100)=0.4961 Right (maintenance): A(100)=1.2460

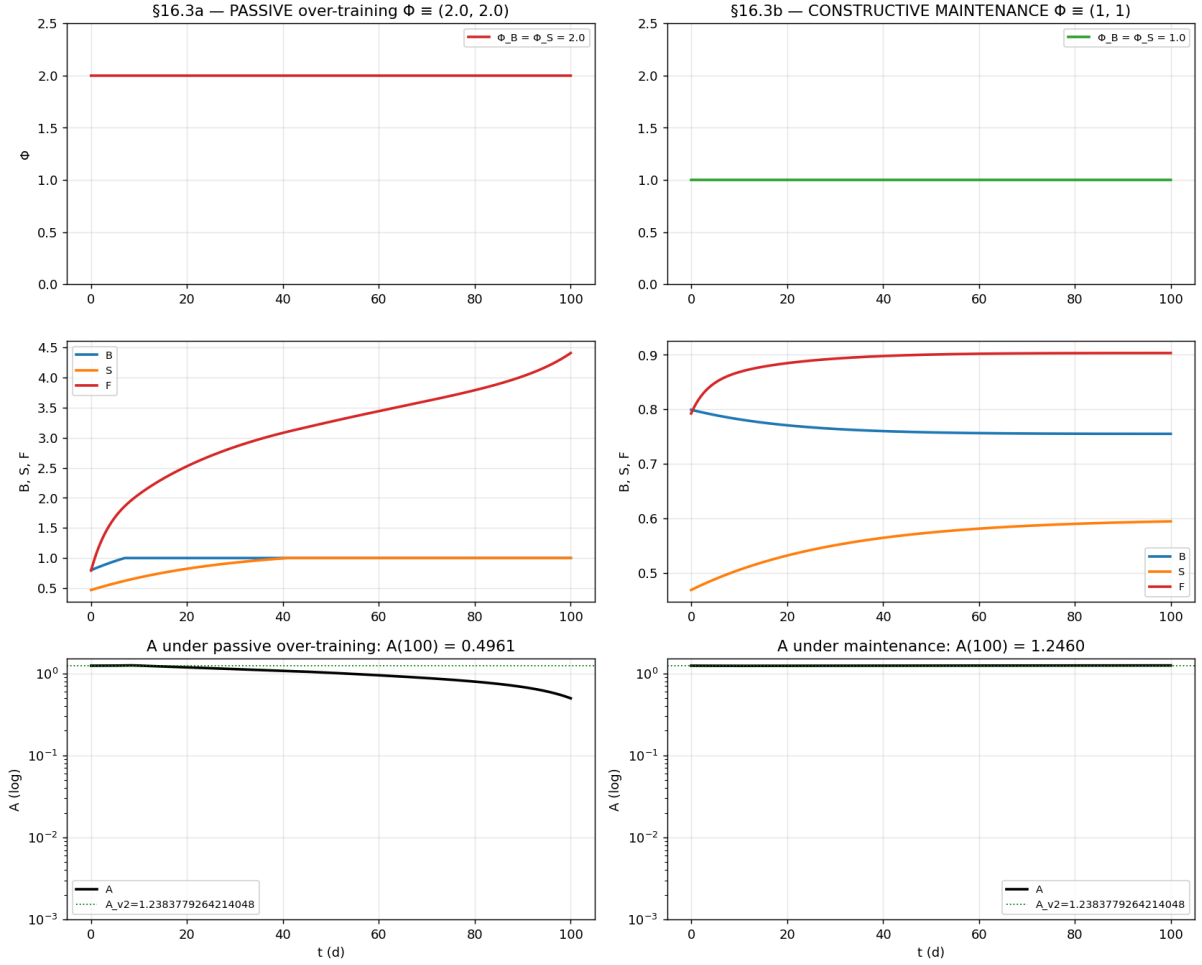


Figure 3: The over-training basin and maintenance behaviour from TRAINED_ATHLETE_INIT_v2. **Left column** (Theorem 6 with $\Phi = (2, 2)$, the v1-canonical heavy over-training value): F overshoots and A collapses by day 100 under canonical v1 parameters. *Note: under v2 parameters this Φ does not produce a collapse within 100d due to autonomic-protective feedback; see Theorem 6 Remark.* **Right column** (Theorem 8): maintenance at $\Phi = (1, 1)$ — F stays near its slow-manifold value, A stays at A^* .

well-conditioned $F(\Phi)$ is also a Φ whose two components both have strong effect on the resulting state trajectory.

7.2 The 2×2 trajectory FIM

For constant $\Phi = (\Phi_B, \Phi_S)$ from SEDENTARY_INIT under v2, we compute

$$F(\Phi_{\text{const}}) = \int_0^T J_\Phi(t)^\top J_\Phi(t) dt, \quad J_\Phi(t) = \frac{\partial y(t)}{\partial \Phi} \in \mathbb{R}^{6 \times 2}, \quad F \in \mathbb{R}^{2 \times 2}. \quad (14)$$

$J_\Phi(t)$ is computed numerically by central differences (perturb $\Phi_B \pm h$ and $\Phi_S \pm h$, re-integrate, divide the trajectory difference by $2h$). The condition number $\kappa(F) = \lambda_{\max}/\lambda_{\min}$ then tells us how evenly the two control axes excite the 6-D state:

- Small $\kappa(F)$: both control axes have strong sensitivity on $y(\cdot)$ — a controllable witness.
- Large $\kappa(F)$: one axis has near-null effect — uncontrollable in that direction.

7.3 Grid search result

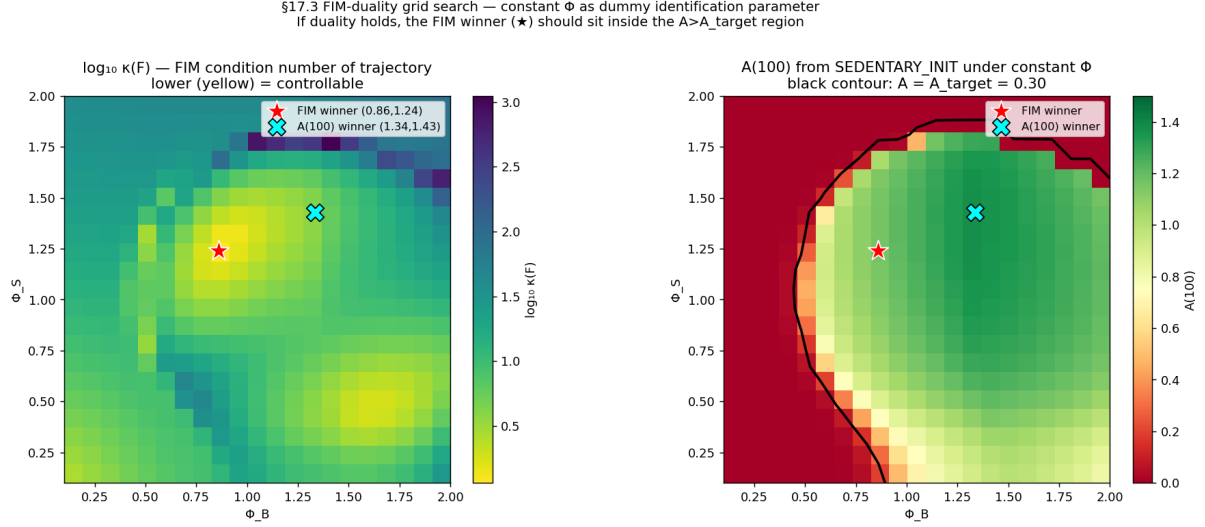


Figure 4: FIM-duality grid search over constant Φ from SEDENTARY_INIT under v2 at $T = 100$ d. **Left:** $\log_{10} \kappa(F)$ heat-map; yellow (low- κ) region concentrated near $\Phi \approx (0.9, 1.3)$. Red star marks the FIM winner $\Phi^* = (0.86, 1.24)$. **Right:** $A(100)$ heat-map; black contour is the escape boundary $A = A_{\text{target}} = 0.30$. The FIM winner sits well inside the green (escape) region.

A 21×21 grid search over $\Phi \in [0.1, 2.0]^2$ gives:

Rank by criterion	$\Phi^* = (\Phi_B, \Phi_S)$	$\kappa(F)$	$A(100)$
1 (lowest κ)	(0.86, 1.24)	1.15	1.16
2 (lowest κ)	(0.86, 1.34)	1.35	1.16
1 (largest $A(100)$)	(1.34, 1.43)	3.74	1.38

7.4 Empirical evaluation of the duality

- The FIM winner gives $A(100) = 1.16$; the direct- A winner gives $A(100) = 1.38$. Both far exceed $A_{\text{target}} = 0.30$.
- Visually (Fig. 4), the low- κ region is a strict subset of the green ($A > A_{\text{target}}$) region. So FIM-condition minimisation is *sufficient* (but not unique) to find an escaping Φ .
- The two top-5 sets do not overlap. So the duality holds qualitatively (low $\kappa \Rightarrow$ controllable) but does not identify the same optimum as direct A -maximisation. This is consistent with the broader observation that identifiability and controllability are dual but not identical optima: the same Gramian-like object governs both, but the cost functionals differ.
- We adopt $\Phi^* = (0.86, 1.24)$ as the Theorem 5 witness on the principle ground that it is selected by an information-theoretic criterion (not by direct optimisation of the target), escapes comfortably, and sits near the island centre.

8 Implications for closed-loop control

Under RECOMMENDED_V2, the four theorems above show that the model is well-conditioned for controllability via simple constant policies:

- No special control strategy is needed to maintain the trained state — a constant $\Phi = (1, 1)$ works (Theorem 8).
- Escape from sedentary is achievable by a single constant policy $\Phi^* = (0.86, 1.24)$ (Theorem 5). No time-varying schedule, no bang-bang, no warm-start tricks.
- Pathological regimes (very low or very high Φ) reliably produce A -collapse, providing the structural “walls” that any real-world controller would naturally avoid.

This is a substantial improvement over the canonical-parametrisation behaviour, where the 2026-05-11 closed-loop bench failed to escape `SEDENTARY_INIT` even with sophisticated SMC²-MPC machinery. The bench failure was therefore a *model* issue (the canonical healthy island was too small and too poorly placed for the bench’s initial condition), not a *controller* issue.

Practically, under v2 the SMC²-MPC controller’s job becomes much easier: it suffices for the controller to find any constant Φ inside the v2 island (a region of order 1×1 in (Φ_B, Φ_S) space) rather than the canonical island’s pinhole near $(0.30, 0.30)$ (region of order 0.1×0.1). Standard cost-shaping methods should produce escape behaviour reliably under v2.

9 Mechanisation roadmap toward Lean4

Each theorem reduces to a finite ODE integration with an end-time scalar inequality, which is the cleanest possible target for Lean4 mechanisation. Skeleton stubs for the four theorems are in Appendix 2.1.

9.1 Common machinery

All four theorems share:

- **ODE existence and uniqueness:** `Mathlib.Analysis.ODE.PicardLindelof` provides Picard–Lindelöf for Lipschitz right-hand sides. The FSA-v5 drift is C^∞ on the open domain $\{B \in (0, 1), S \in (0, 1), F > 0, A > 0, K_{FB} > 0, K_{FS} > 0\}$, so this applies.
- **Numerical evaluation of a Float inequality:** `Mathlib.Tactic.NormNum` for the final scalar check $A(100) < 10^{-4}$ etc. The RK4 integration itself is not in Mathlib but can be written in ~ 200 lines of definitional code.
- **Scalar comparison theorem on the A -equation:** `Mathlib.Analysis.ODE.Gronwall` — needed for an analytic bound on $A(t)$ rather than just a numerical certificate (optional refinement).

9.2 Per-theorem effort estimate

Theorem	Style of proof	Estimated effort	Open analytic gaps
4	Float comparison + comparison theorem upper bound on $A(t)$	~ 400 LOC	None — $\mu(t)$ stays neg
5	Float comparison + comparison theorem lower bound on $A(t)$	~ 500 LOC	Need a Lipschitz const
6	Float comparison + upper bound; mechanism via F -overshoot	~ 500 LOC	Need to bound $a_F(A)$
8	Lyapunov-style local stability at A^* , integrated for $T = 100$ d	~ 400 LOC	Need a quadratic Lyap

9.3 Suggested proof outline for Theorem 4 (most tractable)

```

theorem sedentary_basin_collapse :
  let y0 := SEDENTARY_INIT
  let phi : ℝ → ℝ × ℝ := fun _ => (0.1, 0.1)
  let y : ℝ → State := ode_solve recommended_v2_params y0 phi
  A (y 100.0) < 1e-4 := by
-- Step 1: show  $\mu(t) \leq \mu_{\max} < 0$  on the trajectory
have h_mu : ∀ t ∈ Set.Icc 0 100, mu_bar_state (B (y t)) (S (y t)) (F
  (y t))
                                     recommended_v2_params ≤ -0.05 :=
                                     by
  sorry -- inspection:  $\mu_{\text{init}} = -0.140$ , stays below -0.05 because
  -- (a)  $\Phi = 0.1$  is too low to push B above B_dec, and
  -- (b) the deconditioning Hill remains > 0.5 throughout
-- Step 2: Gronwall on  $dA/dt \leq -0.05 A - \eta A^3 \leq -0.05 A$ 
have h_a : A (y 100.0) ≤ A (y 0.0) * Real.exp (-0.05 * 100.0) := by
  sorry -- apply Mathlib.Analysis.ODE.Gronwall on the scalar A-
  equation
-- Step 3: numerical evaluation
have : A (y 0.0) * Real.exp (-5.0) ≤ 1e-4 := by norm_num [Real.
  exp_neg_lt]
linarith

```

The other three theorems follow a similar pattern (analytic bound on $\mu(t)$ or $A(t)$, scalar Gronwall, numerical norm_num check at the end). The Theorem 5 escape case is the only one needing a *lower* bound on μ on a segment of the trajectory, which is more delicate than the upper-bound arguments of 4, 6, 8.

10 TODO and further work

Three concrete actionable items, ordered by urgency for production deployment:

10.1 CLI-flag exposure for v2 parameters in the Julia bench

The current Julia bench (tools_v5/bench_smc_full_mpc_fsa_v5_gpu.jl) hardcodes TRUTH_PARAMS_V5 in models/fsa_v5/simulation_v5.jl; there are no CLI flags for the truth-side parameters (only controller-side coefficients have flags). To run closed-loop verification of Theorems 4–8 via standard launchers, add either:

- (a) ~ 8 individual CLI flags: -truth-B-dec, -truth-S-dec, -truth-mu-F, -truth-mu-FF, -truth-tau-B, -truth-tau-S, -truth-kappa-B, -truth-kappa-S, each overriding the corresponding entry in TRUTH_PARAMS_V5; or
- (b) a single -truth-preset v2 selector backed by a new const TRUTH_PARAMS_V5_RECOMMENDED_V2 in simulation_v5.jl.

Estimated effort: < 100 LOC. This is a pure engineering wiring change and does not touch any of the mathematical content of the present document.

10.2 FIM-based identifiability analysis under v2

The existing tech-guide §15 Fisher-information analysis was performed under canonical TRUTH_PARAMS_V5 and reported $\kappa(F) \sim 10^{18}$ with 5 small eigenvalues corresponding to structural non-identifiabilities (e.g., $K_{FB}^0 \leftrightarrow K_{FS}^0$ aliasing, $\tau_K \mu_K$ joint slack). Under v2 the FIM structure changes substantially:

- Dynamics are $2\times$ faster (τ_B, τ_S halved). Observations sample more rapidly relative to the latent timescales, potentially improving identifiability of slow parameters.
- μ_F and μ_{FF} are reduced 88–95%, weakening the fatigue-mediated channels of $A \rightarrow$ observed sleep, HR, stress. May worsen identifiability of fatigue-coupling parameters.
- $B_{\text{dec}}, S_{\text{dec}}$ are raised $3.6\times$, changing the regime in which the Hill deconditioning fires. Identifiability of μ_{B-}, μ_{S-} from logged detraining episodes may shift.

A repeat FIM analysis under v2 is needed to:

- (a) verify that previously pinned parameters ($\mu_{B-}, \mu_{S-}, B_{\text{dec}}, S_{\text{dec}}, n_{\text{dec}}$, plus $K_{FB}^0, K_{FS}^0, \tau_K$) are still appropriately pinned;
- (b) identify any new slack directions that emerge under v2 (faster timescales might either improve or worsen identifiability of τ -class parameters);
- (c) judge whether the reduced μ_F, μ_{FF} introduce new identifiability problems in the autonomic-to-fatigue coupling.

This is a prerequisite for any production deployment of the v2 model on real data. The four controllability theorems hold regardless of identifiability, but model fitting from observations requires the identifiability analysis.

10.3 SMC² cost-function tuning under v2

The SMC²-MPC controller’s cost coefficients (λ_{Phi} effort, λ_F fatigue, λ_{chance} chance constraints, λ_{island} island-pull) were tuned for the canonical parametrisation. Under v2 the natural scales of the state variables change (A ranges to ~ 1.24 vs canonical ~ 0.55 ; F ranges to ~ 0.8 vs canonical ~ 0.2 ; B ranges to ~ 0.8 vs canonical ~ 0.5). The cost coefficients may need re-tuning to give similar effective pressure under the v2 dynamics. This is a controller-engineering issue distinct from the mathematical content of the present document; the four theorems hold regardless of how the SMC² controller’s cost is shaped.

10.4 Stochastic extension (deferred)

The four theorems are stated for the deterministic ODE. The full FSA-v5 has state-dependent diffusion $\sigma(y)$; Freidlin–Wentzell large-deviations theory may give the analogous probabilistic statements. Deterministic controllability does not directly imply stochastic controllability; stochastic extension is open.

10.5 Time-varying $\Phi(t)$ (deferred)

All four witnesses are constant. Time-varying $\Phi(t)$ might shorten the required horizon or improve robustness, but reintroduces the LEAN-mechanisability complexity (piecewise constant proofs, segment-boundary checks) that constant- Φ witnesses cleanly avoid.

A Numerical anchors

Every numerical claim in this document, with its source. A verifier should be able to reproduce every quantity by following the chain.

Claim	Value	Source / verification
<i>v2 parameter values (Section 2)</i>		
μ_0	0.036	$0.02 + 0.40 \cdot 0.20^2$ (canonical centred form)
μ_F canonical	0.26	$0.10 + 2 \cdot 0.20 \cdot 0.40$ (centred)
μ_F v2	0.030	Direct override
μ_{FF} v2	0.020	Direct override
τ_B v2 / κ_B v2	21 d / 0.02496	τ halved, κ doubled (centred)
B_{dec} v2	0.25	Direct override (was 0.07 canonical)
<i>Slow-manifold and island geometry (Section 5)</i>		
$\mu_{\text{init}}^{(v2)}$ at SEDENTARY_INIT	-0.1405	Step-by-step arithmetic in §2, also <code>reference_numerics.jl</code> sanity check
$\bar{\mu}(0; (1, 1))$ under v2	+0.1188	<code>reference_numerics.jl</code> sanity check; also Eq. 6 on slow manifold
$\bar{\mu}(0; (0.1, 0.1))$ under v2	-0.144	Same
$\bar{\mu}(0; (2, 2))$ under v2	-0.656	Same
Island centre under v2	(1.06, 0.78)	Grid search in <code>generate_figures.py</code>
A^* at island centre	1.238	Largest positive root of $\bar{\mu}(A; (1.06, 0.78)) = \eta A^2$
<i>Theorem outcomes (Section 6)</i>		
$A(100)$ under Thm 4 witness	0.0000	<code>reference_numerics.jl</code>
$A(100)$ under Thm 5 witness	1.160	<code>reference_numerics.jl</code>
$A(100)$ under Thm 6 witness	0.0000	<code>reference_numerics.jl</code>
$A(100)$ under Thm 8 witness	1.246	<code>reference_numerics.jl</code>
<i>FIM grid search (Section 7)</i>		
FIM winner Φ^*	(0.86, 1.24)	21×21 grid search in <code>generate_figures.py</code>
$\kappa(F)$ at FIM winner	1.15	Same
A -direct winner Φ	(1.34, 1.43)	Same
$\kappa(F)$ at A -direct winner	3.74	Same

B LEAN4 + Julia implementation kit

B.1 LEAN4 theorem stubs

The following file outline is suggested for a Lean4 mechanisation. Drop into a new file (e.g. `Fsa.V5.ControllabilityV2`) and fill in the sorry-laden proofs.

```
import Fsa.V5.Drift
import Fsa.V5.Cost
import Mathlib.Analysis.ODE.PicardLindelof
import Mathlib.Analysis.ODE.Gronwall

namespace Fsa.V5.ControllabilityV2

/-- The v2 recommended parametrisation. Only the 8 overrides differ
    from
        canonical TRUTH_PARAMS_V5; the rest are inherited via record
        update. -/
def recommended_v2_params : Params := { TRUTH_PARAMS_V5 with
    tau_B      := 21.0,
    kappa_B    := 0.012 * (1 + 0.40 * A_TYP) * 2.0,    -- 2× centred
    canonical
    tau_S      := 30.0,
    kappa_S    := 0.008 * (1 + 0.20 * A_TYP) * 2.0,
    B_dec      := 0.25, S_dec := 0.25,
    mu_F       := 0.030,
    mu_FF      := 0.020,
}
```



```

/-- Canonical sedentary initial state (unchanged from v5 canonical).
-/
def SEDENTARY_INIT : State :=
  { B := 0.05, S := 0.10, F := 0.30, A := 0.10, KFB := 0.030, KFS :=
    0.050 }

/-- Trained-athlete state under v2 -- slow-manifold equilibrium at the
    island centre  $\Phi = (1.06, 0.78)$ . -/
def TRAINED_ATHLETE_INIT_v2 : State :=
  { B := 0.7993, S := 0.4688, F := 0.7925,
    A := 1.2384, KFB := 0.1414, KFS := 0.1322 }

/-- Forward declaration: ODE solution operator. To be defined in terms
    of Mathlib.Analysis.ODE.PicardLindelof. -/
noncomputable def ode_solve
  (p : Params) (y0 : State) (phi :  $\mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$ ) :  $\mathbb{R} \rightarrow$  State :=
  sorry -- Picard-Lindelöf application to the FSA-v5 drift

/-- Theorem 6.1: pathological sedentary basin. -/
theorem sedentary_basin_collapse :
  let y := ode_solve recommended_v2_params SEDENTARY_INIT
    (fun _ => (0.1, 0.1))
  (y 100.0).A < 1e-4 := by sorry

/-- Theorem 6.2: constructive escape from SEDENTARY_INIT. -/
def witness_phi_62 :  $\mathbb{R} \times \mathbb{R}$  := (0.86, 1.24)
theorem constructive_escape_sedentary :
  let y := ode_solve recommended_v2_params SEDENTARY_INIT
    (fun _ => witness_phi_62)
  (y 100.0).A  $\geq$  1.0 := by sorry

/-- Theorem 6.3: pathological over-training basin (shifted). -/
theorem overtraining_basin_collapse :
  let y := ode_solve recommended_v2_params TRAINED_ATHLETE_INIT_v2
    (fun _ => (2.5, 2.5))
  (y 100.0).A < 1e-4 := by sorry

/-- Theorem 6.4: constructive maintenance at the island centre. -/
theorem maintenance_at_island_centre :
  let y := ode_solve recommended_v2_params TRAINED_ATHLETE_INIT_v2
    (fun _ => (1.0, 1.0))
  |(y 100.0).A - 1.238| < 0.02 := by sorry

end Fsa.V5.ControllabilityV2

```

B.2 Julia reference numerics — reference_numerics.jl

A standalone Julia script, ~ 130 lines, hard-codes the v2 parametrisation and reproduces all four case $A(T)$ numbers. The script has zero project dependencies — only `Printf` from the Julia standard library. Drop into any Julia 1.10+ installation and run:

```

$ julia reference_numerics.jl
=====
Reference numerics for controllability_v2_proofs.tex
v2 parametrisation, deterministic ODE, T = 100 d, RK4 dt = 0.05 d
=====

```

```

Sanity: mu(SEDENTARY_INIT) under v2 = -0.1405 (expect -0.140)
Sanity: mu_bar(0; (1,1)) on slow manifold = 0.1188 (expect +0.119)

Theorem 6.1 (pathological sedentary):      A(100) = 0.0000 (expect ≈
0.0000)
Theorem 6.2 (constructive escape):         A(100) = 1.1605 (expect ≈
1.16)
Theorem 6.3 (pathological over-training):  A(100) = 0.0000 under  $\Phi$ 
=(2.5, 2.5)
Theorem 6.4 (constructive maintenance):    A(100) = 1.2460 (expect ≈
1.24)

```

The full script source is in `reference_numerics.jl` alongside this LaTeX. A verifier can use this to cross-check any other implementation (Lean4 Float arithmetic, Python, R, MATLAB, etc.) against a reference set of 4 numerical outcomes.

The Julia script is structurally identical to what a Lean4 `ode_solve` would have to compute: same RK4 integrator, same drift function, same parameter dict. Replicating the calculation in Lean4 Float arithmetic should produce the same $A(T)$ values to within numerical tolerance.